

# Manuscript outline

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## Introduction

1. The most frequently observed biological impact of climate change is major changes in phenology<sup>1,2,3,4,5</sup>
  - (a) Shifts in spring and autumn phenology modifies when seasons start and end<sup>6</sup>.
  - (b) Understanding the consequences of these shifts requires understanding how much and why growing seasons have changed<sup>6</sup>.
  - (c) Spring phenology advances,<sup>7,8,9</sup> driven by temperature<sup>10,2,11</sup>
  - (d) Autumn phenology delays, but much less than spring<sup>12,13</sup> and driven by temperature and photoperiod<sup>14,15,16,17</sup>
2. Shifts in spring and autumn phenology supports the assumption that longer growing seasons increase tree growth<sup>18,19</sup>
  - (a) Recent research shows that this may not be the case<sup>20,21,22,23 24,25,23</sup>
  - (b) <sup>20</sup> showed that trees did not grow more or faster despite warmer and earlier springs at the regional scale
  - (c) ? showed a decoupling between GPP and carbon storage at the global scale.
  - (d) If trees don't grow as we expect, it could affect forest carbon cycle projections<sup>26,27</sup> and subsequent climate solutions<sup>21</sup>
3. Understanding these finds requires answering why trees do not grow more despite longer growing seasons.
  - (a) We have a poor understanding of carbon allocation to above-ground biomass, despite it being one of the largest carbon sink<sup>24</sup>.
  - (b) Linear representation of growth as a function of carbon assimilation is wrong<sup>24</sup>
  - (c) Trees may be source or sink limited, which emphasizes the importance of understanding the temperature sensitivity relationship between growth and photosynthesis<sup>28</sup>.
  - (d) Growth may be more sensitive to varying environmental conditions than photosynthesis, which could mislead models<sup>24,29,30,31</sup>
  - (e) Hard nature of climate change makes understanding the internal physiological and external constraints to growth critical<sup>32,24,33</sup>
4. How trees internal programming and their susceptibility to environmental conditions could constrain their growth?
  - (a) Growth as any other biological mechanism is genetically controlled, and will thus vary by species<sup>34,33?</sup>
  - (b) Tree plasticity is a highly variable trait that changes with species' life history strategies<sup>35,36</sup>, affecting how flexible their growth is in response to varying environmental conditions.

- (c) While CO<sub>2</sub> concentrations increased, drought, heat waves, competition, insect outbreaks and spring frosts—factors projected to change in frequency and severity with a warming climate—can offset the benefits of longer seasons<sup>37,38,39,40,41,42,43,44</sup>.
  - (d) Droughts can act as a severe growth constraint<sup>45</sup> by stopping growth early<sup>46</sup> or inducing tissue mortality<sup>39</sup>.
  - (e) Intra- and inter-specific competition for nutrients, water and light can also offset a tree's capacity to take advantage of extra days in the season<sup>47</sup>.
5. Understanding the constraints to growth requires defining the growing season and growth.
- (a) Different definitions of GS: we propose to use calendar (number of days) and thermal season (window of opportunity of thermally favorable meteorological conditions for plant growth)<sup>48</sup>
  - (b) The calendar season only accounts for the number of days in the season as the the predictors for growth, implying a stable growth rate throughout the season.
  - (c) This may reveal a role of some neglected cooler and shorter days in the season that have a weak influence on the thermal season.
  - (d) Growth rate changes with temperature, which the thermal season accounts for and should therefore better predict growth responses.
  - (e) Wood formation (xylogenesis) is the allocation of carbon for long-term storage<sup>49,50</sup>, and the radial growth increments are represented through tree rings<sup>51</sup>/
  - (f) Allometries using tree DBH and height have a coarse temporal scale and can miss extreme events affecting growth e.g.,<sup>52,53</sup>.
  - (g) Tree rings are more precise growth proxies, which we use here<sup>54,55</sup>
6. Growth drivers differences across species need to be considered.
- (a) Phenology and the relationship between growth and growing season length varies heavily across species<sup>49</sup>
  - (b) Pooling all species together in carbon sequestrations models is a weakness<sup>21,24,33</sup>
  - (c) We propose to use ground-based leaf phenology observations across several species to address this issue<sup>56</sup>
  - (d) These observations provide valuable insights of the growth temporality of trees not suitable for experimental trials<sup>33</sup>
  - (e) Cambial phenology in diffuse-porous trees is closely synchronized with leaf phenology<sup>57,50,58</sup>, so we can use them to as a reliable proxy for start and end of the growing season<sup>19</sup>
7. Local adaptation could inform growth responses of species under future climate change<sup>59,33</sup>.
- (a) Provenance refers to the geographic and climatic origin of a population<sup>60,61</sup>
  - (b) Trees from northern provenances sometimes have later spring phenology<sup>62</sup> and often end of season autumn phenology=shorter growing season<sup>63</sup>.
  - (c) This is especially true for budset—which marks the end of primary growth—, with its phenotypic variance being strongly related to provenance and photoperiod acting as the primary cue<sup>60,59,64</sup>.
  - (d) Thus, common gardens with provenances across a latitudinal gradient can inform tree's growth responses to future climatic conditions.
8. To answer if trees grow more with longer growing seasons, we related three years of leaf phenology and growth data across four species and four provenances in a common garden on juvenile trees.
- (a) Determine if trees grow more with longer thermal vs calendar season (ref: conceptual figure)
  - (b) Understand the calendar growing season relationship with growth and how it relates to the SOS and EOS
  - (c) Determine how growth changes across species and their provenances

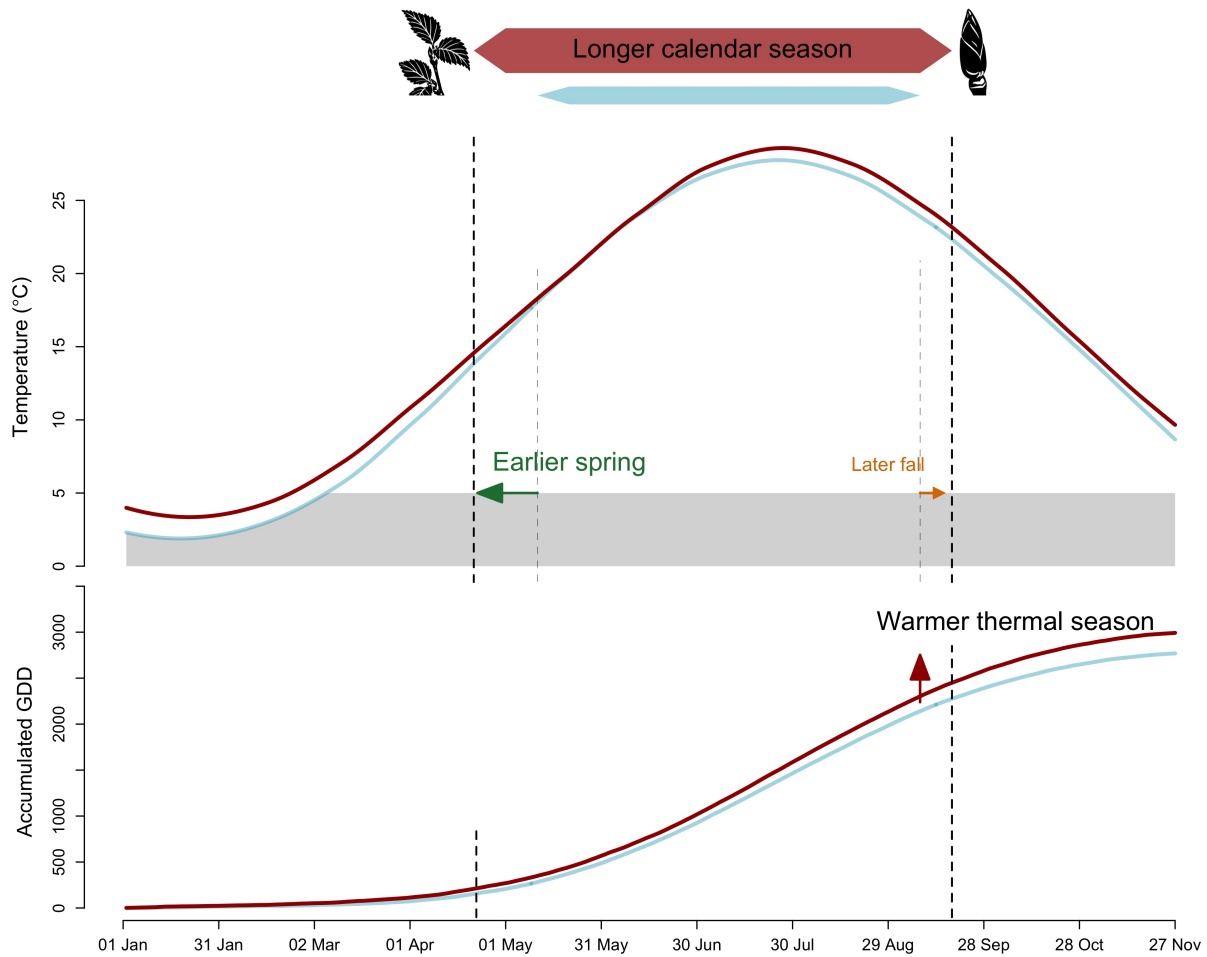


Figure 1: We show tree ring widths (mm) across years and species in two ways: (a) as model estimates for the year intercepts with the dots depicting the mean and the lines, the 50% and 90% uncertainty intervals (UI) and (b) using box plots of the empirical data. Though the trend is weak, 2020 is systematically the year with the lowest growth shown both as the mean of the model intercepts (a) and as the median of the raw data, for each species (b). The growth of 2018 and 2019 is similar.

## Methods

## Results

Separate document

## Discussion

1. What we did, how we did it and what we found
  - (a) We used 81 juvenile trees of four species across four provenances to understand if longer growing seasons using leaf phenology increased growth using tree rings
  - (b) We found that longer growing seasons increased growth for all species, but varied depending on the metric

- (c) Thermal season was a better predictor for growth than the calendar season
- 2. Thermal season being a better predictor than the calendar season may be caused by it weighting the relative importance of the days on growth better than the calendar season
  - (a) Growth rate follows a temperature response curve, thus trees don't grow at the same speed all year round
  - (b) Cool and short spring days count for less in the thermal season compared to the calendar season where each day counts the same
- 3. Disentangling the calendar season with the start and end of season may reveal something...
  - (a) 123
- 4. Our trees grew more with longer seasons in absence of external constraints
  - (a) Droughts can strongly impact tree growth, thus when seasons are longer, but not drier, trees just may be able to fully take advantage of these extra days
  - (b) Earlier spring means trees start water uptake earlier, which may increase the severity of droughts later in the season
  - (c) Dow showed that growth and growing season length are decoupled in a natural system where trees compete for water, but also for light and nutrients
  - (d) Though we did not control for nutrients, our study design allowed us to control for light availability where taller trees did not shade the others at the life stage we looked at
  - (e) The timing of phenological phases are shifting and this will affect inter-specific competition for resources, this highlights the importance of accounting for competition when addressing this question
- 5. Our species have different life history strategies and this may inform how they respond to growing season length
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- 6. We found little effect of provenance, suggesting local adaption on growth may be weak under current conditions
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- 7. Our study has limitations
  - (a) We had low sample size and not individuals for each provenance for all species
  - (b) We didn't account for extreme temperatures in our models, though there was no extended heat wave periods throughout the study
- 8. Our results highlights the importance of accounting for
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## References

1. Parmesan, C. and Yohe, G. A globally coherent fingerprint of climate change impacts across natural systems. *Nature* **421**(6918), 37–42, January (2003).
2. Cleland, E., Chuine, I., Menzel, A., Mooney, H., and Schwartz, M. Shifting plant phenology in response to global change. *Trends in Ecology & Evolution* **22**(7), 357–365, July (2007).
3. Lieth, H., Jacobs, J., Lange, O. L., Olson, J. S., and Wieser, W., editors. *Phenology and Seasonality Modeling*, volume 8 of *Ecological Studies*. Springer Berlin Heidelberg, Berlin, Heidelberg, (1974).

4. Woolway, R. I., Sharma, S., Weyhenmeyer, G. A., Debolskiy, A., Golub, M., Mercado-Bettín, D., Perroud, M., Stepanenko, V., Tan, Z., Grant, L., Ladwig, R., Mesman, J., Moore, T. N., Shatwell, T., Vanderkelen, I., Austin, J. A., DeGasperi, C. L., Dokulil, M., La Fuente, S., Mackay, E. B., Schladow, S. G., Watanabe, S., Marcé, R., Pierson, D. C., Thiery, W., and Jennings, E. Phenological shifts in lake stratification under climate change. *Nature Communications* **12**(1), 2318, April (2021).
5. Menzel, A., Sparks, T. H., Estrella, N., Koch, E., Aasa, A., Ahas, R., Alm-Kübler, K., Bissolli, P., Braslavská, O., Briede, A., Chmielewski, F. M., Crepinsek, Z., Curnel, Y., Defila, C., Donnelly, A., Filella, Y., Jatzcak, K., Mestre, A., Peñuelas, J., Pirinen, P., Scheifinger, H., Striz, M., Susnik, A., Van Vliet, A. J. H., Wielgolaski, F., Zach, S., and Züst, A. European phenological response to climate change matches the warming pattern. *Global Change Biology* **12**(10), 1969–1976, October (2006).
6. Duputié, A., Rutschmann, A., Ronce, O., and Chuine, I. Phenological plasticity will not help all species adapt to climate change. *Global Change Biology* **21**(8), 3062–3073, August (2015).
7. Wolfe, D. W., Schwartz, M. D., Lakso, A. N., Otsuki, Y., Pool, R. M., and Shaulis, N. J. Climate change and shifts in spring phenology of three horticultural woody perennials in northeastern USA. *International Journal of Biometeorology* **49**(5), 303–309, May (2005).
8. Chmielewski, F.-M. and Rötzer, T. Response of tree phenology to climate change across Europe. *Agricultural and Forest Meteorology* **108**(2), 101–112, June (2001).
9. Fu, Y. H., Piao, S., Op De Beeck, M., Cong, N., Zhao, H., Zhang, Y., Menzel, A., and Janssens, I. A. Recent spring phenology shifts in western Central Europe based on multiscale observations. *Global Ecology and Biogeography* **23**(11), 1255–1263, November (2014).
10. Chuine, I. Why does phenology drive species distribution? *Philosophical Transactions of the Royal Society B: Biological Sciences* **365**(1555), 3149–3160, October (2010).
11. Peñuelas, J. and Filella, I. Responses to a Warming World. *Science* **294**(5543), 793–795, October (2001).
12. Gallinat, A. S., Primack, R. B., and Wagner, D. L. Autumn, the neglected season in climate change research. *Trends in Ecology & Evolution* **30**(3), 169–176, March (2015).
13. Jeong, S. and Medvigy, D. Macroscale prediction of autumn leaf coloration throughout the continental United States. *Global Ecology and Biogeography* **23**(11), 1245–1254, November (2014).
14. Cooke, J. E. K., Eriksson, M. E., and Junttila, O. The dynamic nature of bud dormancy in trees: environmental control and molecular mechanisms. *Plant, Cell & Environment* **35**(10), 1707–1728, October (2012).
15. Flynn, D. F. B. and Wolkovich, E. M. Temperature and photoperiod drive spring phenology across all species in a temperate forest community. *New Phytologist* **219**(4), 1353–1362, September (2018).
16. Körner, C. and Basler, D. Phenology Under Global Warming. *Science* **327**(5972), 1461–1462, March (2010).
17. Delpierre, N., Vitasse, Y., Chuine, I., Guillemot, J., Bazot, S., Rutishauser, T., and Rathgeber, C. B. K. Temperate and boreal forest tree phenology: from organ-scale processes to terrestrial ecosystem models. *Annals of Forest Science* **73**(1), 5–25, March (2016).
18. Keenan, T. F., Gray, J., Friedl, M. A., Toomey, M., Bohrer, G., Hollinger, D. Y., Munger, J. W., O’Keefe, J., Schmid, H. P., Wing, I. S., Yang, B., and Richardson, A. D. Net carbon uptake has increased through warming-induced changes in temperate forest phenology. *Nature Climate Change* **4**(7), 598–604, July (2014).
19. Stridbeck, P., Björklund, J., Fuentes, M., Gunnarson, B. E., Jönsson, A. M., Linderholm, H. W., Ljungqvist, F. C., Olsson, C., Rayner, D., Rocha, E., Zhang, P., and Seftigen, K. Partly decoupled tree-ring width and leaf phenology response to 20th century temperature change in Sweden. *Dendrochronologia* **75**, 125993, October (2022).

20. Dow, C., Kim, A. Y., D'Orangeville, L., Gonzalez-Akre, E. B., Helcoski, R., Herrmann, V., Harley, G. L., Maxwell, J. T., McGregor, I. R., McShea, W. J., McMahon, S. M., Pederson, N., Tepley, A. J., and Anderson-Teixeira, K. J. Warm springs alter timing but not total growth of temperate deciduous trees. *Nature* **608**(7923), 552–557, August (2022).
21. Green, J. K. and Keenan, T. F. The limits of forest carbon sequestration. *Science* **376**(6594), 692–693, May (2022).
22. Silvestro, R., Zeng, Q., Buttò, V., Sylvain, J.-D., Drolet, G., Mencuccini, M., Thiffault, N., Yuan, S., and Rossi, S. A longer wood growing season does not lead to higher carbon sequestration. *Scientific Reports* **13**(1), 4059, March (2023).
23. Wang, W., Qin, L., Zhang, T., Yang, F., Cabon, A., Wang, Z., Zhou, P., Zhang, Y., Fonti, P., and Huang, J. Seasonal climate variations drive decoupling between the duration and amount of xylem growth along a hydrothermal gradient in the southern Altai Mountains. *Journal of Ecology* **113**(7), 1793–1806, July (2025).
24. Cabon, A., Kannenberg, S. A., Arain, A., Babst, F., Baldocchi, D., Belmecheri, S., Delpierre, N., Guerrieri, R., Maxwell, J. T., McKenzie, S., Meinzer, F. C., Moore, D. J. P., Pappas, C., Rocha, A. V., Szejner, P., Ueyama, M., Ulrich, D., Vincke, C., Voelker, S. L., Wei, J., Woodruff, D., and Anderegg, W. R. L. Cross-biome synthesis of source versus sink limits to tree growth. *Science* **376**(6594), 758–761, May (2022).
25. Matula, R., Knířová, S., Vítámvás, J., Šrámek, M., Kníř, T., Ulbrichová, I., Svoboda, M., and Plichta, R. Shifts in intra-annual growth dynamics drive a decline in productivity of temperate trees in Central European forest under warmer climate. *Science of The Total Environment* **905**, 166906, December (2023).
26. Richardson, A. D., Keenan, T. F., Migliavacca, M., Ryu, Y., Sonnentag, O., and Toomey, M. Climate change, phenology, and phenological control of vegetation feedbacks to the climate system. *Agricultural and Forest Meteorology* **169**, 156–173, February (2013).
27. Swidrak, I., Schuster, R., and Oberhuber, W. Comparing growth phenology of co-occurring deciduous and evergreen conifers exposed to drought. *Flora: Morphology, Distribution, Functional Ecology of Plants* **208**(10-12), 609–617, December (2013).
28. Gessler, A. and Zweifel, R. Beyond source and sink control – toward an integrated approach to understand the carbon balance in plants. *New Phytologist* **242**(3), 858–869, May (2024).
29. Cabon, A., Fernández-de-Uña, L., Gea-Izquierdo, G., Meinzer, F. C., Woodruff, D. R., Martínez-Vilalta, J., and De Cáceres, M. Water potential control of turgor-driven tracheid enlargement in Scots pine at its xeric distribution edge. *New Phytologist* **225**(1), 209–221, January (2020).
30. Muller, B., Pantin, F., Génard, M., Turc, O., Freixes, S., Piques, M., and Gibon, Y. Water deficits uncouple growth from photosynthesis, increase C content, and modify the relationships between C and growth in sink organs. *Journal of Experimental Botany* **62**(6), 1715–1729, March (2011).
31. Peters, R. L., Steppe, K., Cuny, H. E., De Pauw, D. J., Frank, D. C., Schaub, M., Rathgeber, C. B., Cabon, A., and Fonti, P. Turgor – a limiting factor for radial growth in mature conifers along an elevational gradient. *New Phytologist* **229**(1), 213–229, January (2021).
32. Almagro, D., Martin-Benito, D., Rossi, S., Conde, M., Fernández-de-Uña, L., and Gea-Izquierdo, G. Long-Term Cambial Phenology Reveals Diverging Growth Responses of Two Tree Species in a Mixed Forest Under Climate Change. *Global Change Biology* **31**(9), e70503, September (2025).
33. Wolkovich, E. M., Ettinger, A. K., Chin, A. R., Chamberlain, C. J., Baumgarten, F., Pradhan, K., Manzanedo, R. D., and Hille Ris Lambers, J. Why longer seasons with climate change may not increase tree growth. *Nature Climate Change* **15**(12), 1283–1292, December (2025).

34. Howe, G. T., Aitken, S. N., Neale, D. B., Jermstad, K. D., Wheeler, N. C., and Chen, T. H. From genotype to phenotype: unraveling the complexities of cold adaptation in forest trees. *Canadian Journal of Botany* **81**(12), 1247–1266, December (2003).
35. Cuny, H. E., Fonti, P., Rathgeber, C. B., Von Arx, G., Peters, R. L., and Frank, D. C. Couplings in cell differentiation kinetics mitigate air temperature influence on conifer wood anatomy. *Plant, Cell & Environment* **42**(4), 1222–1232, April (2019).
36. Michelot, A., Simard, S., Rathgeber, C., Dufrene, E., and Damesin, C. Comparing the intra-annual wood formation of three European species (*Fagus sylvatica*, *Quercus petraea* and *Pinus sylvestris*) as related to leaf phenology and non-structural carbohydrate dynamics. *Tree Physiology* **32**(8), 1033–1045, August (2012).
37. Tyree, M. T. and Zimmermann, M. H. *Xylem Structure and the Ascent of Sap*. Springer Series in Wood Science. Springer Berlin Heidelberg, Berlin, Heidelberg, (2002).
38. Choat, B., Brodribb, T. J., Brodersen, C. R., Duursma, R. A., López, R., and Medlyn, B. E. Triggers of tree mortality under drought. *Nature* **558**(7711), 531–539, June (2018).
39. Li, Y., Zhang, W., Schwalm, C. R., Gentine, P., Smith, W. K., Ciais, P., Kimball, J. S., Gazol, A., Kannenberg, S. A., Chen, A., Piao, S., Liu, H., Chen, D., and Wu, X. Widespread spring phenology effects on drought recovery of Northern Hemisphere ecosystems. *Nature Climate Change* **13**(2), 182–188, February (2023).
40. Trenberth, K. E., Dai, A., Van Der Schrier, G., Jones, P. D., Barichivich, J., Briffa, K. R., and Sheffield, J. Global warming and changes in drought. *Nature Climate Change* **4**(1), 17–22, January (2014).
41. Change, I. P. O. C. Detection and Attribution of Climate Change: from Global to Regional. In *Climate Change 2013 – The Physical Science Basis*, 867–952. Cambridge University Press 1 edition, March (2014).
42. Chiang, F., Mazdiasni, O., and AghaKouchak, A. Evidence of anthropogenic impacts on global drought frequency, duration, and intensity. *Nature Communications* **12**(1), 2754, May (2021).
43. Polgar, C. A. and Primack, R. B. Leaf-out phenology of temperate woody plants: from trees to ecosystems. *New Phytologist* **191**(4), 926–941, September (2011).
44. Reinmann, A. B., Bowers, J. T., Kaur, P., and Kohler, C. Compensatory responses of leaf physiology reduce effects of spring frost defoliation on temperate forest tree carbon uptake. *Frontiers in Forests and Global Change* **6**, 988233, February (2023).
45. Kang, J., Yang, Z., Yu, B., Ma, Q., Jiang, S., Shishov, V. V., Zhou, P., Huang, J.-G., and Ding, X. An earlier start of growing season can affect tree radial growth through regulating cumulative growth rate. *Agricultural and Forest Meteorology* **342**, 109738, November (2023).
46. Dox, I., Skrøppa, T., Decoster, M., Prislan, P., Gascó, A., Gričar, J., Lange, H., and Campioli, M. Severe drought can delay autumn senescence of silver birch in the current year but advance it in the next year. *Agricultural and Forest Meteorology* **316**, 108879, April (2022).
47. Cleland, E. E. and Wolkovich, E. Effects of Phenology on Plant Community Assembly and Structure. *Annual Review of Ecology, Evolution, and Systematics* **55**(1), 471–492, November (2024).
48. Körner, C., Möhl, P., and Hiltbrunner, E. Four ways to define the growing season. *Ecology Letters* **26**(8), 1277–1292, August (2023).
49. Etzold, S., Sterck, F., Bose, A. K., Braun, S., Buchmann, N., Eugster, W., Gessler, A., Kahmen, A., Peters, R. L., Vitasse, Y., Walthert, L., Ziemińska, K., and Zweifel, R. Number of growth days and not length of the growth period determines radial stem growth of temperate trees. *Ecology Letters* **25**(2), 427–439, February (2022).

50. Silvestro, R., Deslauriers, A., Prislan, P., Rademacher, T., Rezaie, N., Richardson, A. D., Vitasse, Y., and Rossi, S. From Roots to Leaves: Tree Growth Phenology in Forest Ecosystems. *Current Forestry Reports* **11**(1), 12, January (2025).
51. Plomion, C., Leprovost, G., and Stokes, A. Wood Formation in Trees. *Plant Physiology* **127**(4), 1513–1523, December (2001).
52. Meyer, H. A. A Mathematical Expression for Height Curves. *Journal of Forestry* **38**(5), 415–420, May (1940).
53. Saunders, M. R. and Wagner, R. G. Height-diameter models with random coefficients and site variables for tree species of Central Maine. *Annals of Forest Science* **65**(2), 203–203, January (2008).
54. Gazol, A., Camarero, J. J., Vicente-Serrano, S. M., Sánchez-Salguero, R., Gutiérrez, E., De Luis, M., Sangüesa-Barreda, G., Novak, K., Rozas, V., Tíscar, P. A., Linares, J. C., Martín-Hernández, N., Martínez Del Castillo, E., Ribas, M., García-González, I., Silla, F., Camisón, A., Génova, M., Olano, J. M., Longares, L. A., Hevia, A., Tomás-Burguera, M., and Galván, J. D. Forest resilience to drought varies across biomes. *Global Change Biology* **24**(5), 2143–2158, May (2018).
55. Manzanedo, R. D. and Pederson, N. Towards a More Ecological Dendroecology. *Tree-Ring Research* **75**(2), 152, August (2019).
56. Piao, S., Liu, Q., Chen, A., Janssens, I. A., Fu, Y., Dai, J., Liu, L., Lian, X., Shen, M., and Zhu, X. Plant phenology and global climate change: Current progresses and challenges. *Global Change Biology* **25**(6), 1922–1940 (2019).
57. Marchand, L. J., Dox, I., Gričar, J., Prislan, P., Van Den Bulcke, J., Fonti, P., and Campioli, M. Timing of spring xylogenesis in temperate deciduous tree species relates to tree growth characteristics and previous autumn phenology. *Tree Physiology* **41**(7), 1161–1170, July (2021).
58. Suzuki, M., Yoda, K., and Suzuki, H. Phenological Comparison of the Onset of Vessel Formation Between Ring-Porous and Diffuse-Porous Deciduous Trees in a Japanese Temperate Forest. *IAWA Journal* **17**(4), 431–444 (1996).
59. Soolanayakanahally, R. Y., Guy, R. D., Silim, S. N., and Song, M. Timing of photoperiodic competency causes phenological mismatch in balsam poplar ( *Populus balsamifera* L.). *Plant, Cell & Environment* **36**(1), 116–127, January (2013).
60. Aitken, S. N. and Bemmels, J. B. Time to get moving: assisted gene flow of forest trees. *Evolutionary Applications* **9**(1), 271–290, January (2016).
61. Housset, J. M., Nadeau, S., Isabel, N., Depardieu, C., Duchesne, I., Lenz, P., and Girardin, M. P. Tree rings provide a new class of phenotypes for genetic associations that foster insights into adaptation of conifers to climate change. *New Phytologist* **218**(2), 630–645, April (2018).
62. Zeng, Z. A. and Wolkovich, E. M. Weak evidence of provenance effects in spring phenology across Europe and North America. *New Phytologist* **242**(5), 1957–1964, June (2024).
63. Vitasse, Y., eLenz, A., and eKoerner, C. The interaction between freezing tolerance and phenology in temperate deciduous trees. *Frontiers in Plant Science* **5**, October (2014).
64. Way, D. A. and Montgomery, R. A. Photoperiod constraints on tree phenology, performance and migration in a warming world. *Plant, Cell & Environment* **38**(9), 1725–1736 (2015). eprint: <https://onlinelibrary.wiley.com/doi/pdf/10.1111/pce.12431>.



## Supplemental results